Digital Signal Processing

Zhou Hang

Hybrid Time Extraction FFT Algorithm

The length N of the FFT algorithm sequence extracted from base 2 time must satisfy $N=2^M$

The length N of the FFT algorithm sequence extracted from base 3 time must satisfy $N=3^M$

The length N of the FFT algorithm sequence extracted from base 4 time must satisfy $N=4^M$

If N does not meet this requirement, the sequence needs to be zeroed.

The computational complexity of FFT algorithm is directly proportional to the length of the sequence,

Therefore, a large number of zero padding will reduce the actual effectiveness of the FFT algorithm.

Hybrid Time Extraction FFT Algorithm

For example, if the length N of a certain sequence is 96, and FFT is used to calculate its DFT,

Using radix-FFT calculation, it is necessary to add zeros to 128 points (adding 32 zeros)

Using the radix-3-FFT calculation, it is necessary to add zeros to 243 points (adding 147 zeros)

Using the radix-4-FFT calculation, it is necessary to fill in zeros to 256 points (adding 160 zeros)

How to reduce the number of zeros and improve the computational efficiency of FFT?

Hybrid Time Extraction FFT Algorithm

First, decompose the 96 point sequence into three sets of 32 point subsequences, i.e. 96=3×32

Calculate the DFT of these three sets of 32 point subsequences using radix-FFT,

Combine three sets of 32 point DFT according to the rules of radix-3-FFT to form a 96 point sequence DFT.

$$X[m]=X_1[m]+W_N^m X_2[m]+W_N^{2m} X_3[m]$$

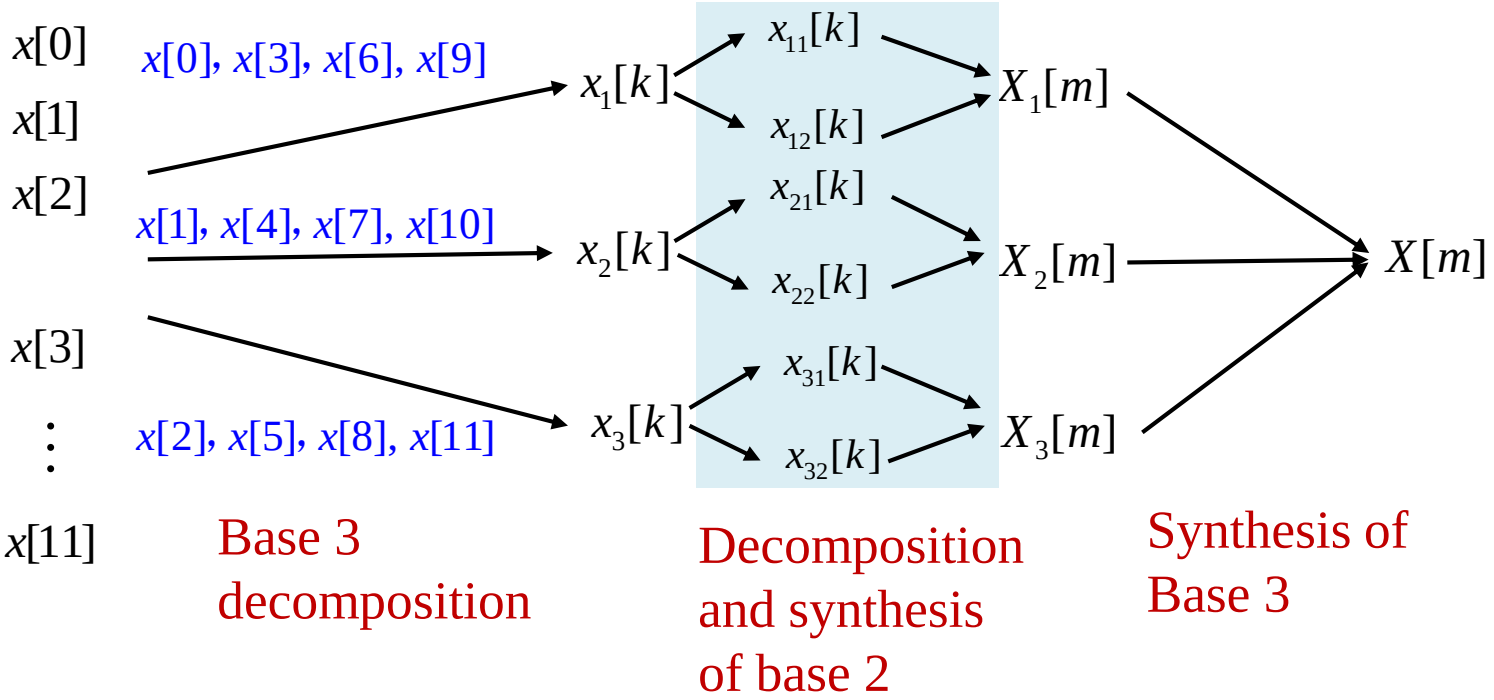
$$X[m+N/3]=X_1[m]+W_3^1 W_N^m X_2[m]+W_3^2 W_N^{2m} X_3[m]$$

$$X[m+2N/3]=X_1[m]+W_3^2 W_N^m X_2[m]+W_3^4 W_N^{2m} X_3[m]$$

Due to the use of FFT with different bases in this calculation process, it is called mixed basis FFT.

E.g: Use the mixed basis FFT algorithm to calculate the DFT of an N=12 point sequence and draw its algorithm flowchart.

N=12 can be expressed as N=3×4. The sequence $x[k]$ is extracted and decomposed into three sets of 4-point sequences according to time, that is, the sequence $x[k]$ is extracted according to the basis 3 method to form three sets of subsequences $\boldsymbol{x_1[k], x_2[k], x_3[k]}$



E.g: Use the mixed basis FFT algorithm to calculate the DFT of an N=12 point sequence and draw its algorithm flowchart.

N=12 can be expressed as N=3×4. The sequence $x[k]$ is extracted and decomposed into three sets of 4-point sequences according to time, that is, the sequence $x[k]$ is extracted according to the basis 3 method to form three sets of subsequences $\boldsymbol{x_1[k], x_2[k], x_3[k]}$

$$x_1[k] = x[3k] = \{x[0], x[3], x[6], x[9]\}$$
$$x_2[k] = x[3k+1] = \{x[1], x[4], x[7], x[10]\} \qquad k=0, 1, 2, 3$$
$$x_3[k] = x[3k+2] = \{x[2], x[5], x[8], x[11]\}$$

The corresponding DFT transform is $X_1[m]$, $X_2[m]$, $X_3[m]$, then the DFT of the long sequence is

$$X[m] = X_1[m] + W_{12}^{m} X_2[m] + W_{12}^{2m} X_3[m]$$
$$X[m+4] = X_1[m] + W_{12}^{(m+4)} X_2[m] + W_{12}^{2(m+4)} X_3[m] \qquad m=0, 1, 2, 3$$
$$X[m+8] = X_1[m] + W_{12}^{(m+8)} X_2[m] + W_{12}^{2(m+8)} X_3[m]$$

E.g: Use the mixed basis FFT algorithm to calculate the DFT of an N=12 point sequence and draw its algorithm flowchart.

$$
\begin{aligned}
X[m] &= X_1[m] + W_{12}^{m} X_2[m] + W_{12}^{2m} X_3[m] \\
X[m+4] &= X_1[m] + W_{12}^{(m+4)} X_2[m] + W_{12}^{2(m+4)} X_3[m] \qquad m=0,1,2,3 \\
X[m+8] &= X_1[m] + W_{12}^{(m+8)} X_2[m] + W_{12}^{2(m+8)} X_3[m]
\end{aligned}
$$

Simplify to obtain:

$$
\begin{aligned}
X[m] &= X_1[m] + W_{12}^{m} X_2[m] + W_{12}^{2m} X_3[m] \\
X[m+4] &= X_1[m] + W_{12}^{4} W_{12}^{m} X_2[m] + W_{12}^{8} W_{12}^{2m} X_3[m] \qquad m=0,1,2,3 \\
X[m+8] &= X_1[m] + W_{12}^{8} W_{12}^{m} X_2[m] + W_{12}^{16} W_{12}^{2m} X_3[m]
\end{aligned}
$$

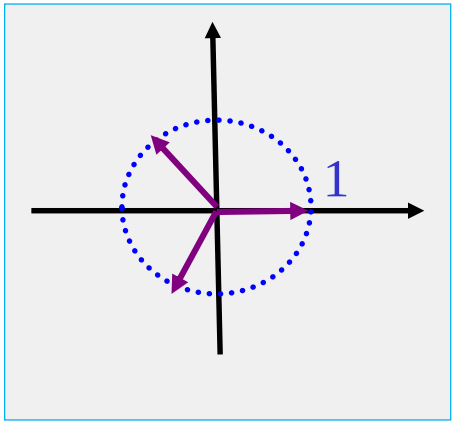
$$
a = W_{12}^{4} = W_{12}^{16} = e^{j\frac{2\pi}{12}\times 4} = -\frac{1}{2} + j\frac{\sqrt{3}}{2} \qquad b = W_{12}^{8} = e^{j\frac{2\pi}{12}\times 8} = e^{j\frac{4\pi}{3}} = -\frac{1}{2} - j\frac{\sqrt{3}}{2}
$$

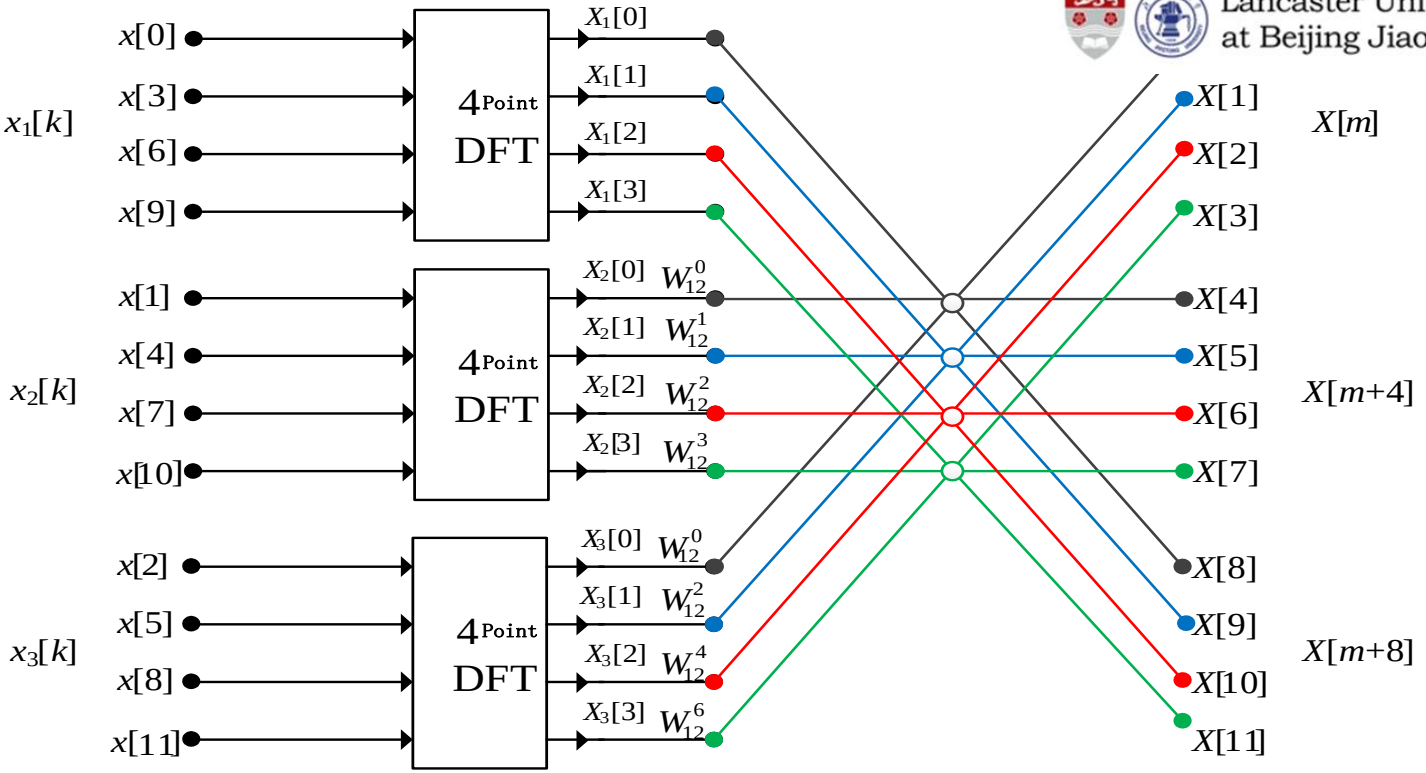
E.g: Use the mixed basis FFT algorithm to calculate the DFT of an N=12 point sequence and draw its algorithm flowchart.

$$X[m] = X_1[m] + W_{12}^{m} X_2[m] + W_{12}^{2m} X_3[m]$$

$$X[m+4] = X_1[m] + aW_{12}^{m} X_2[m] + bW_{12}^{2m} X_3[m] \qquad m = 0, 1, 2, 3$$

$$X[m+8] = X_1[m] + bW_{12}^{m} X_2[m] + aW_{12}^{2m} X_3[m]$$





Base 3 decomposition Synthesis of Base 3

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.